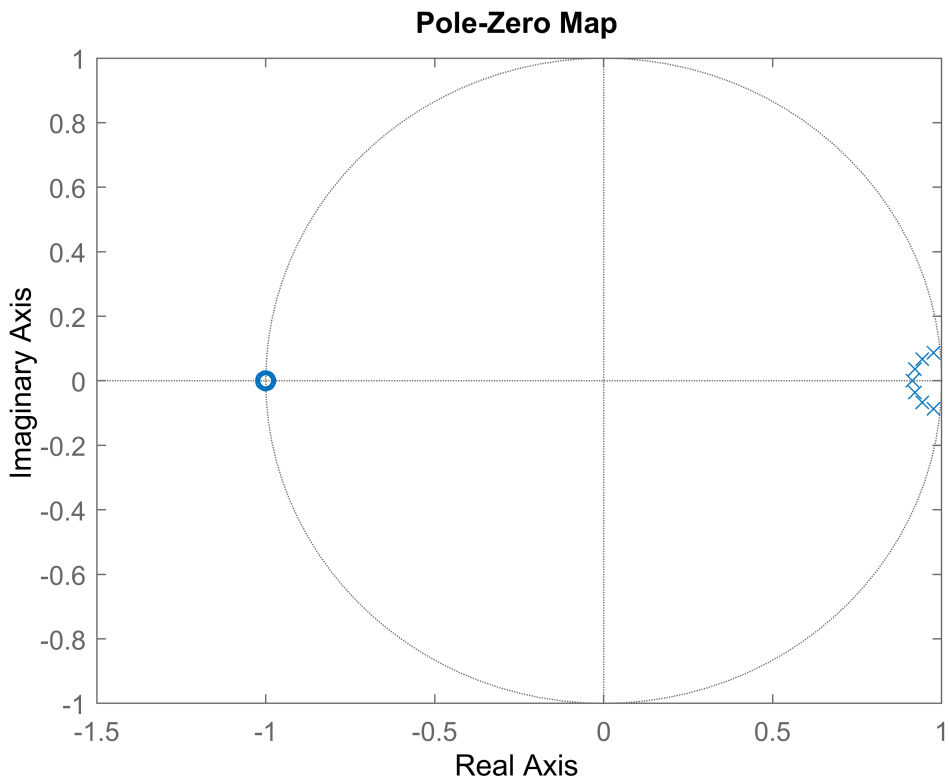
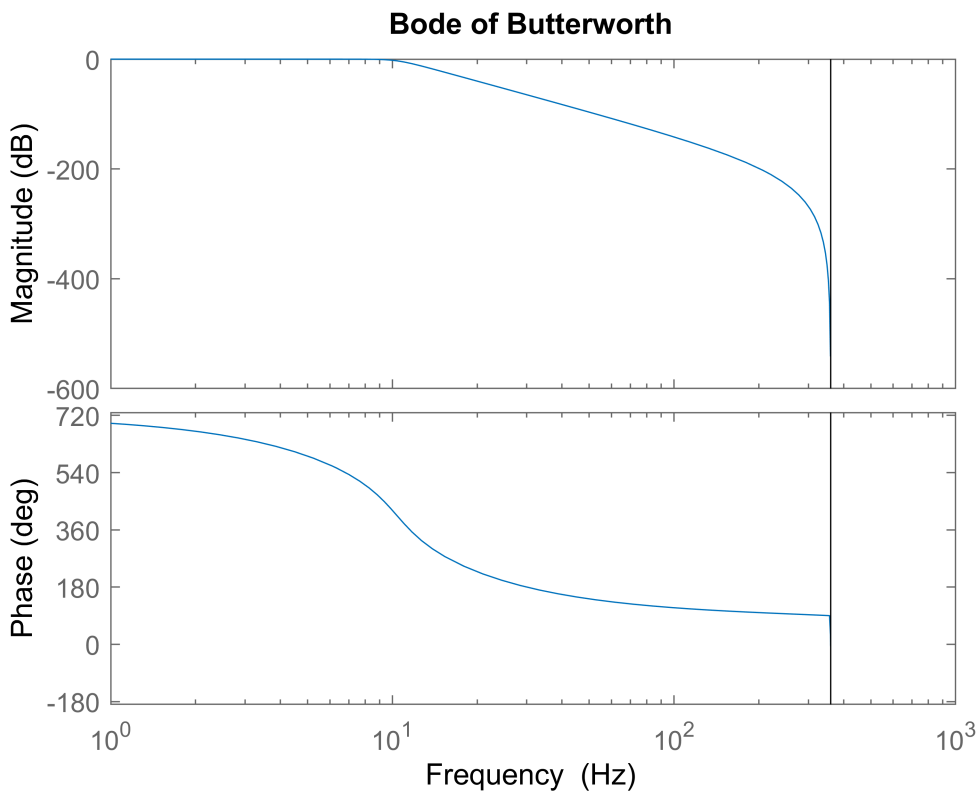
```
% a = 3
Fs = 720;
% Finding order and Natural frequency
[N,wn] = buttord(10*2/Fs,20*2/Fs,3,40);
% finding the numerator and denominator coefficients of transfer function
[B,A] = butter(N,wn);
% finding the TF
SYS = tf(B,A,1/Fs);

% plotting poles and zeroes
figure();
pzmap(SYS);
```




```
% plotting bode
figure();
h = bodeplot(SYS);
setoptions(h,'FreqUnits','Hz');
title('Bode of Butterworth');
```

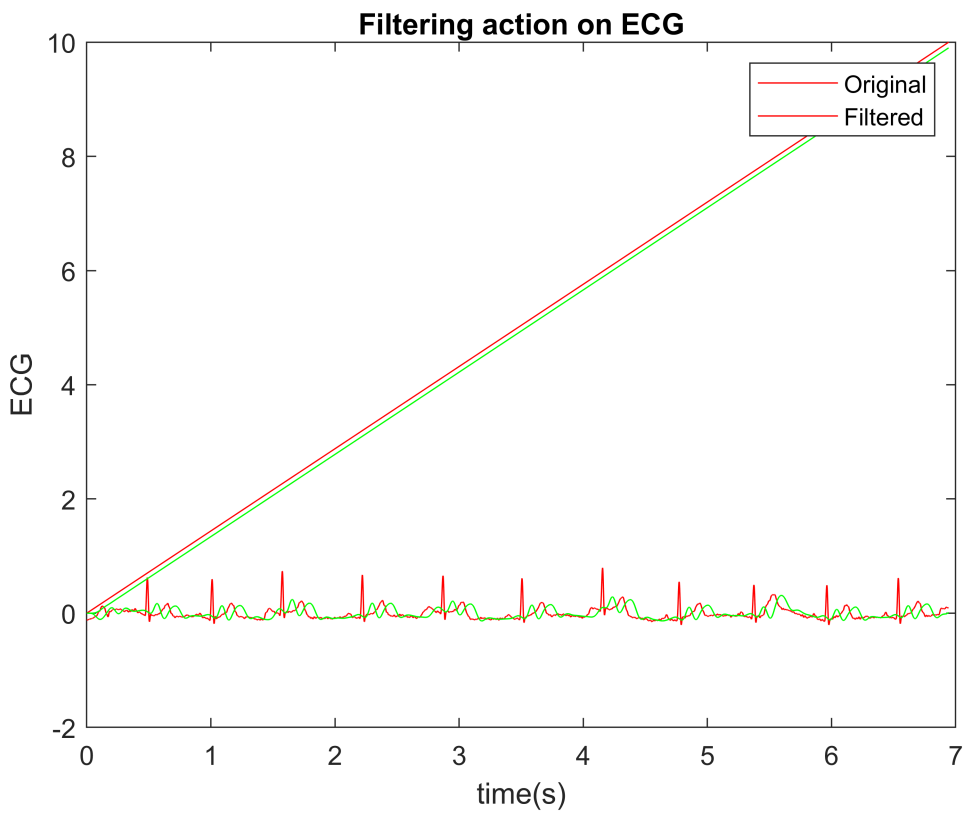



```
% a = 3
Fs = 720;

% Finding order and Natural frequency
[N,wn] = buttord(10*2/Fs,20*2/Fs,3,40);
% finding the numerator and denominator coefficients of transfer function
[B,A] = butter(N,wn);

% loading ECG data
ECG = load('C:\Users\student.SSPU\Downloads\ECG_Data.txt');
% filtering ECG data
Y = filter(B,A,ECG);

dt = 1/Fs;
n = length(ECG);
T = n/Fs;
t = 0:dt:T-dt;

% plotting the filtered ECG signal
figure();
plot(t,ECG,'r');
hold();
```

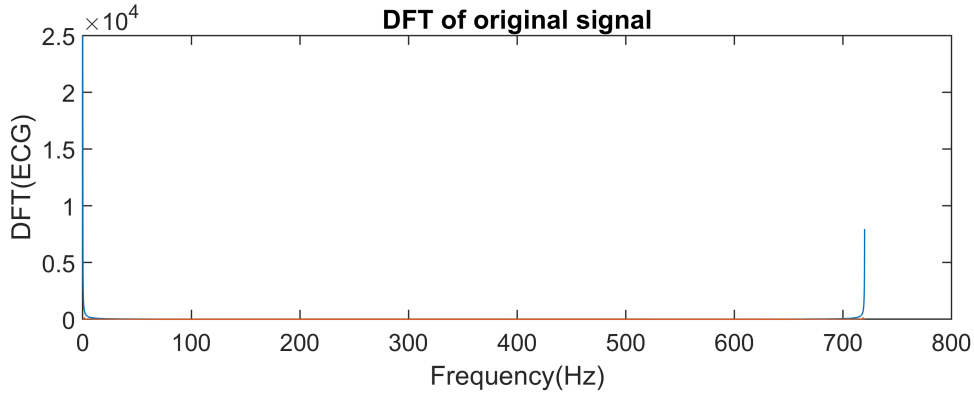
```
Current plot held
```

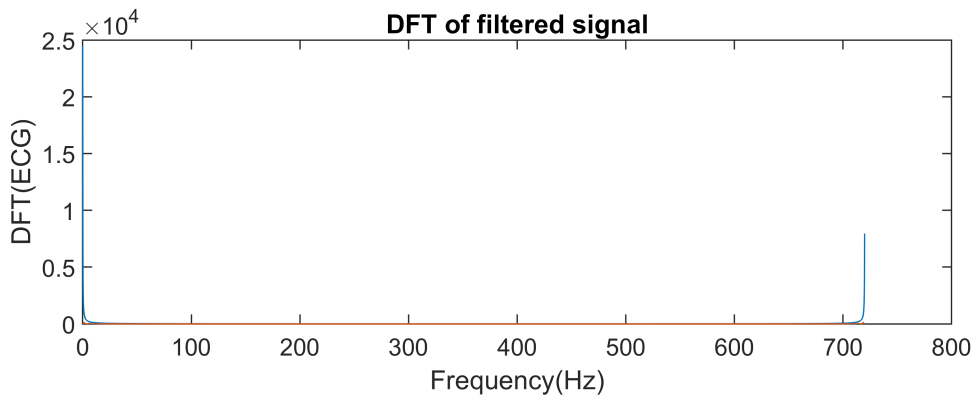
```
plot(t,Y,'g');
legend(['Original';'Filtered']);
```

```
title('Filtering action on ECG');
xlabel('time(s)');
ylabel('ECG');
```

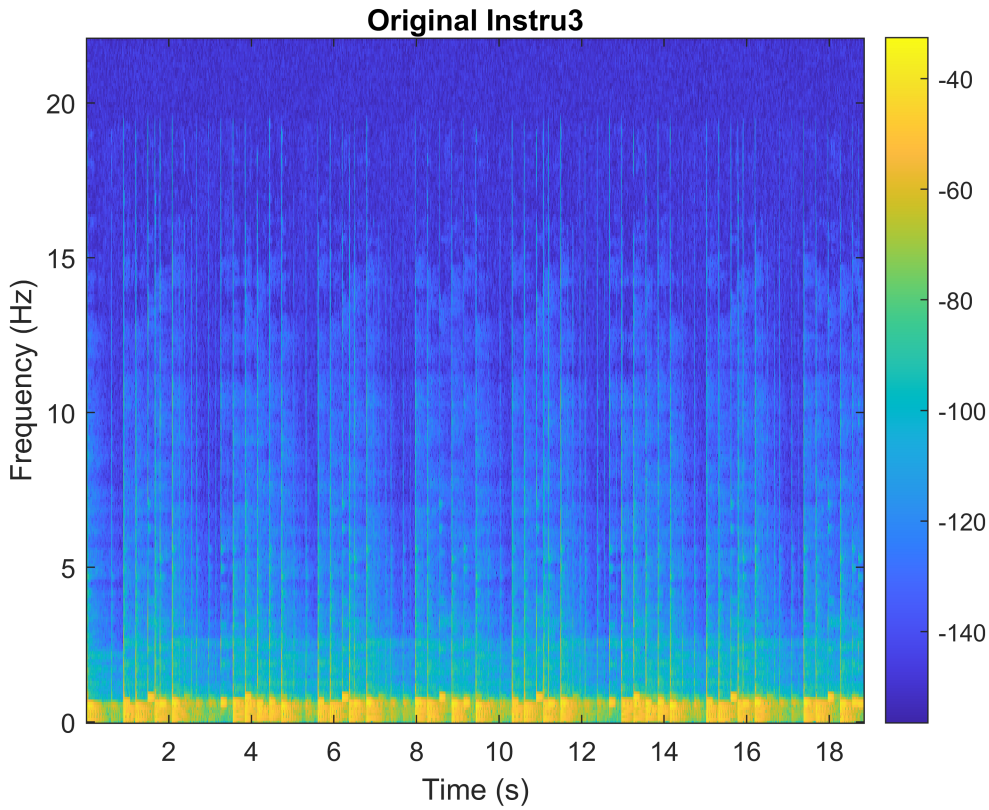



```
DFTx = fft(ECG);
DFTy = fft(Y);
N = length(DFTx);
k = 0:N-1;
f = Fs/N .* k;

% plotting the DFT
figure();
subplot(2,1,1);
plot(f,abs(DFTx));
title('DFT of original signal');
xlabel('Frequency(Hz)');
ylabel('DFT(ECG)');
subplot(2,1,2);
plot(f,abs(DFTy));
title('DFT of filtered signal');
xlabel('Frequency(Hz)');
ylabel('DFT(ECG)');
```

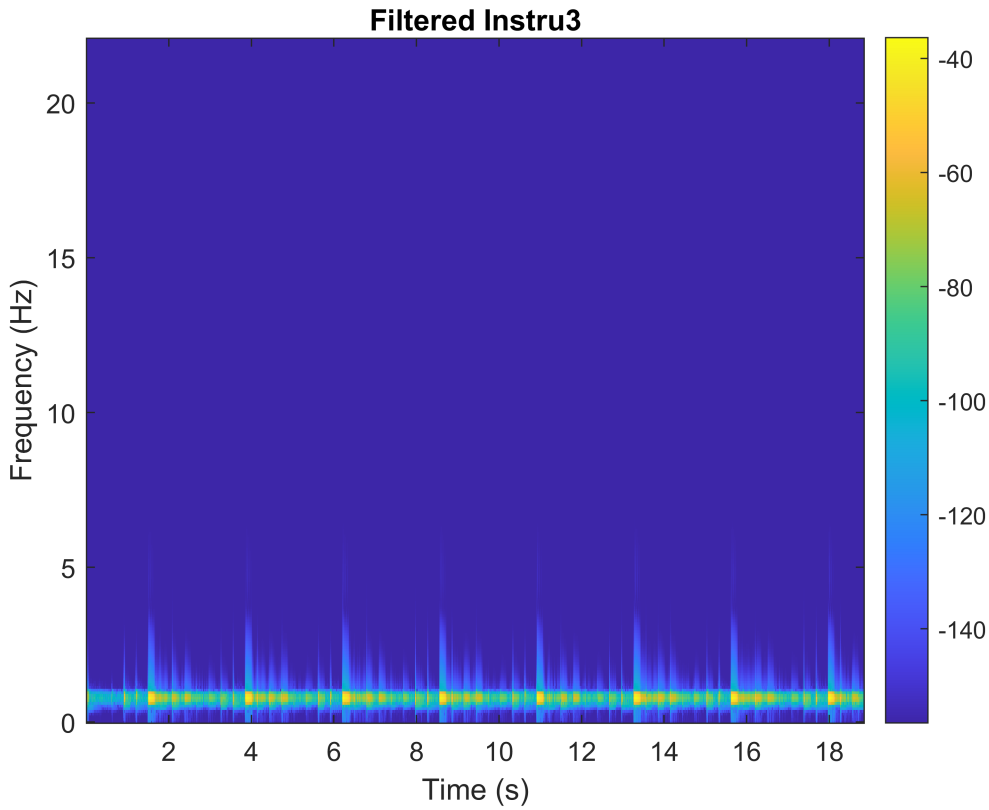




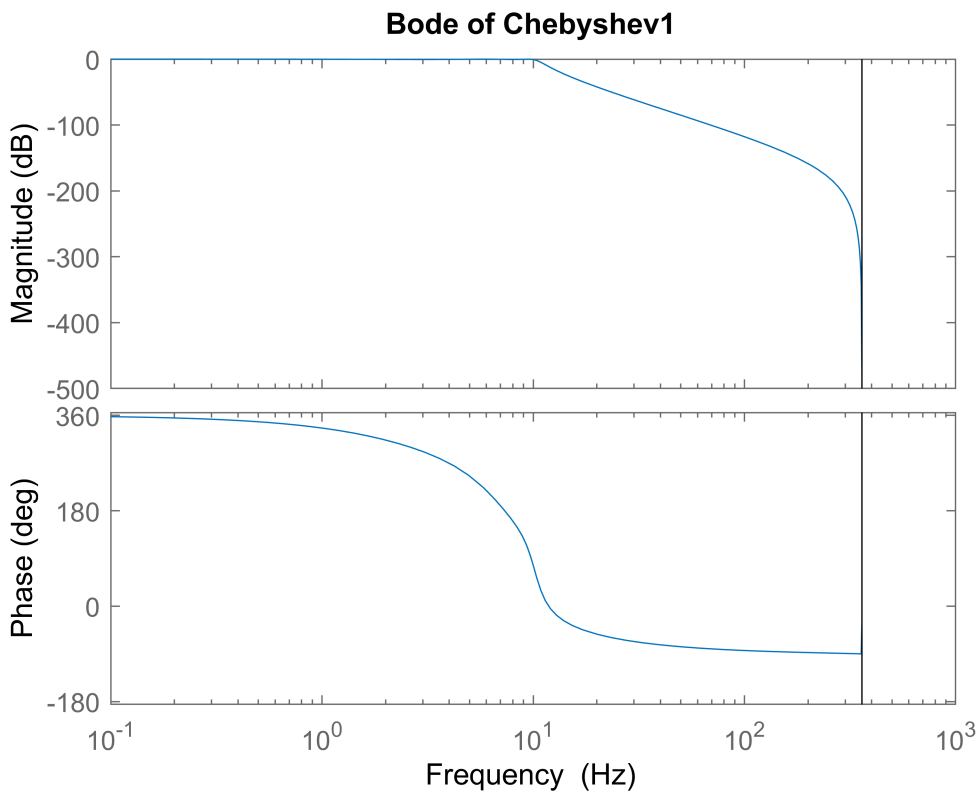

```
% importing the wav file
[y,Fs] = audioread('C:\Users\student.SSPU\Downloads\instru3.wav');

% Check if the audio data is stereo and convert to mono if necessary
if size(y, 2) == 2
y = mean(y, 2); % Convert to mono by averaging the channels
end
% Plotting the spectrogram of the original signal
figure;
spectrogram(y, blackman(500), 250, [], Fs, 'yaxis');
title('Original Instru3');
xlabel('Time (s)');
ylabel('Frequency (Hz)');
colorbar;
```



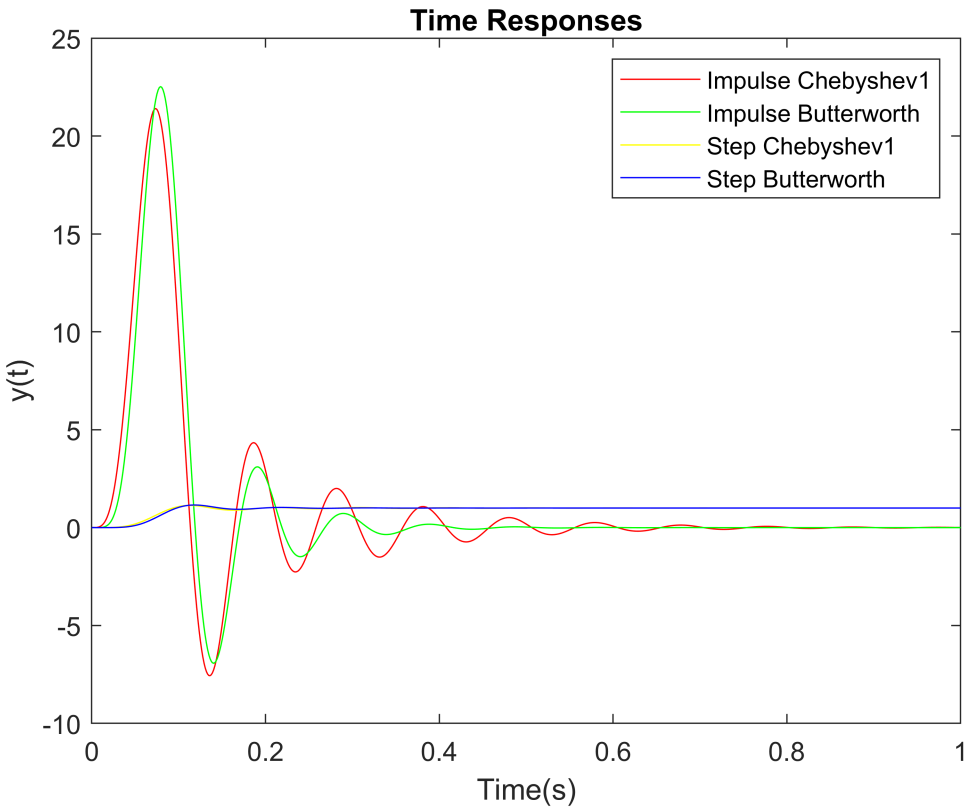

```
% Designing the Butterworth filter
passband = [700, 850] * 2 / Fs; % Normalized passband
stopband = [690, 950] * 2 / Fs; % Normalized stopband
% Calculate the filter order and cutoff frequencies
[N, Wn] = buttord(passband, stopband, 10, 15);
[B, A] = butter(N, Wn);
% Filtering the audio signal
Y = filter(B, A, y);
% Writing the filtered audio to a new file
audiowrite('filtered_instru3.wav', Y, Fs);
% Plotting the spectrogram of the filtered signal
figure;
spectrogram(Y, blackman(500), 250, [], Fs, 'yaxis');
title('Filtered Instru3');
xlabel('Time (s)');
ylabel('Frequency (Hz)');
colorbar;
```




```
%Q4)
% a = 3
Fs = 720;
% designing chebyshev1 filter
[N,wp] = cheb1ord(10*2/Fs,20*2/Fs,3,40);
[B,A] = cheby1(N,0.5,wp);
SYS1 = tf(B,A,1/Fs);
% plotting bode of chebyshev1
figure();
h = bodeplot(SYS1);
setoptions(h,'FreqUnits','Hz');
title('Bode of Chebyshev1');
```




```
% designing butterworth filter
[N,wn] = buttord(10*2/Fs,20*2/Fs,3,40);
[B,A] = butter(N,wn);
SYS2 = tf(B,A,1/Fs);
% finding impulse and step responses of chebyshev1 and butterworth filters
[Y1,T1] = impulse(SYS1,1);
[Y2,T2] = step(SYS1,1);
[Y3,T3] = impulse(SYS2,1);
[Y4,T4] = step(SYS2,1);
% plotting the impulse and step responses
figure();
plot(T1,Y1,'r',T3,Y3,'g',T2,Y2,'y',T4,Y4,'b');
legend('Impulse Chebyshev1','Impulse Butterworth','Step Chebyshev1','Step Butterworth');
title('Time Responses');
xlabel('Time(s)');
ylabel('y(t)');
```


Time Responses
Impulse Chebyshev1
Impulse Butterworth
Step Chebyshev1
Step Butterworth
y(t)
Time(s)
25
20
15
10
5
0
-5
-10
0
0.2
0.4
0.6
0.8
1